# Unions in C

A **Union** is a special user-defined data type in C that allows you to store different data types in the same memory location. While it looks similar to a structure, its internal memory management is completely different.

> **CodeHive Insight:** Think of a union as a single storage box that can hold a tool, a toy, or a book—but only one of them at any given time.

# 1. Union Definition

We use the `union` keyword. Note that the syntax is identical to `struct`, but the behavior is unique.

```
union Data {
    int i;
    float f;
    char c;
};
```

This definition tells the compiler that variables of type `union Data` can hold either an integer, a float, or a character.

## 2. Memory Management

This is the most critical part of Unions. Unlike structures, the total memory allocated for a union is equal to the **size of its largest member**.

Example: If `int` is 4 bytes and `double` is 8 bytes, the union will be exactly 8 bytes.

```
printf("Size: %zu", sizeof(union Data)); // Outputs size of largest member
```

# 3. Accessing Members

You access members using the dot (`.`) operator. However, remember the **Golden Rule**: only one member holds a valid value at a time.

```
union Data d;
d.i = 10;
printf("%d", d.i); // Valid

d.f = 5.5;
printf("%f", d.f); // Valid, but d.i is now corrupted!
```

# 4. Structure vs. Union

| Feature | Structure | Union |
|---|---|---|
| Memory | Separate for each member | Shared by all members |
| Storage | Stores multiple values | Stores one at a time |
| Size | Sum of member sizes | Max of member sizes |

## 5. When to Use Unions? ⭐

- **Memory Savings:** In embedded systems with very little RAM (kilobytes).
- **Variant Types:** When a variable needs to change its data type depending on the situation.
- **Hardware Registers:** For accessing specific bits or bytes of a larger data value.

### Key Points (Exam Ready ✅)

• **union** keyword is mandatory.

• It is highly **memory efficient**.

• Overwriting one member affects all other members.

• Commonly used in low-level systems programming.

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